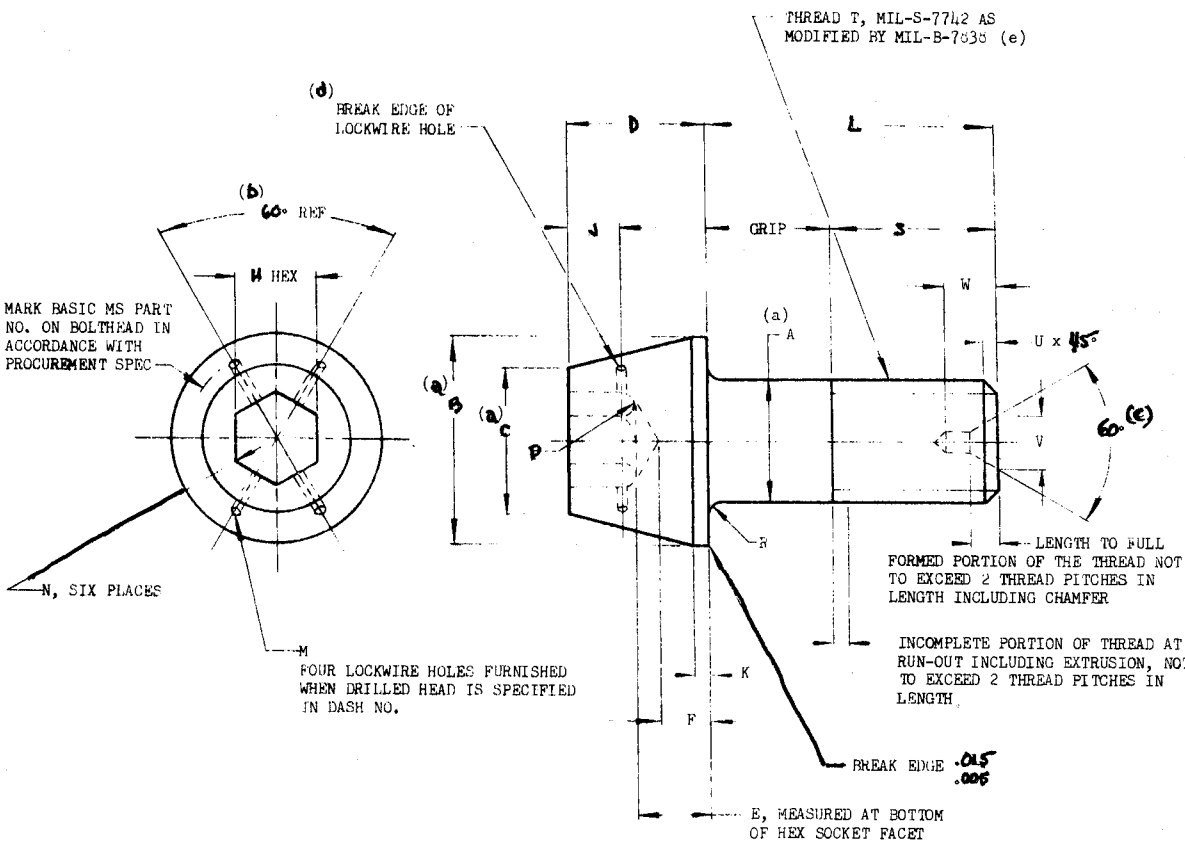
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(a) DIMENSIONS A, B, AND C SHALL BE CONCENTRIC TO EACH OTHER WITHIN VALUES SPECIFIED FOR X TOTAL INDICATOR READING.
DIMENSIONS H AND C SHALL BE CONCENTRIC TO EACH OTHER WITHIN VALUES SPECIFIED FOR Y TOTAL INDICATOR READING.
DIMENSION A AND THREAD PITCH DIAMETER SHALL BE CONCENTRIC TO EACH OTHER WITHIN VALUES SPECIFIED FOR Z TOTAL INDICATOR READING.
(b) REFERENCE DIMENSIONS ARE FOR DESIGN PURPOSES AND TO PROVIDE INFORMATION ONLY. THESE DIMENSIONS ARE NOT INSPECTION OR MANUFACTURING REQUIREMENTS. S MIN REF = (L MIN) - (GRIP MAX).
(c) ON MS20012 THRU MS20024 CENTER DRILLING OF SHANK IS OPTIONAL FOR BOLT MANUFACTURER.
(d) COUNTERSINKING OF DRILLED HOLES IS OPTIONAL.
(e) 1-14 NS INACTIVE FOR DESIGN.

NOTES:
1. MATERIAL: ALLOY STEEL. SEE PROCUREMENT SPECIFICATION.
2. FINISH: CADMIUM PLATING PER QQ-P-416, TYPE II, CLASS 2. PARTS WITH CLASS 3 PLATING MAY BE FURNISHED FROM SUPPLIER'S STOCK UNTIL 1 JANUARY 1975.
3. HARDNESS: ROCKWELL C34 TO C40.
4. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.010, ANGLES ±1°.
5. DIMENSIONS TO BE MET AFTER PLATING.

INTERCHANGEABILITY RELATIONSHIP: BOLTS WITH TYPE II PLATING CAN BE USED TO REPLACE BOLTS WITH TYPE I PLATING IN ALL APPLICATIONS.
BOLTS WITH TYPE I PLATING CAN BE USED IN PLACE OF BOLTS WITH TYPE II PLATING UNTIL PRESENT STOCKS ARE DEPLETED, BUT BOLTS WITH TYPE I PLATING SHALL NOT BE USED AFTER 1 JANUARY 1964.

FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.

(D) DENOTES CHANGE

APPROVED 28 Jan 52 REVISED (A) 20 Apr 62 (B) 10 Dec 62 (C) 29 DEC 72 (D) 10 August 1981

USER SYMBOLS: ARMY -AV

REVIEWER SYMBOLS: ARMY - ER ME
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P.A. Navy - AS Other Cust Army - AR USAF - 99	TITLE BOLTS, INTERNAL WRENCHING, 160 KSI FTU AND 96 KSI FSU	MILITARY STANDARD MS 20004 THRU 20024
PROCUREMENT SPECIFICATION MIL-B-7838	SUPERSEDES:	SHEET 1 OF 4

DD FORM 1 SEP 67 672-1 (Coordinated)

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TABLE I BOLT DIMENSIONS AND TENSILE STRENGTH

BASIC MS PART NO.	(a) THREAD T	(a) A DIA MAX	(a) A DIA MIN	(a) B DIA MAX	(a) B DIA MIN	(a) C DIA	D	E	F MIN	(a) H HEX MAX	(a) H HEX MIN	J	K	M +.005 -.000 DIA	N MAX RAD
MS20004	1/4 -28 UNF-3A	.2492	.2477	.438	.428	.342	.250	.1 2	.068	.1900	.1880	.078		.037	
MS20005	5/16-24 UNF-3A	.3117	.3102	.531	.521	.394	.312	.1 4	.094	.2210	.2190	.094			.010
MS20006	3/8 -24 UNF-3A	.3742	.3727	.649	.639	.520	.375	. 03	.116			.125	.063		
MS20007	7/16-20 UNF-3A	.4367	.4347	.750	.740	.534	.438	.208	.141	.3150	.3130	.156			
MS20008	1/2 -20 UNF-3A	.4991	.4971	.828	.818	.625	.500	.247	.169	.3785	.3755	.172			.011
MS20009	9/16-18 UNF-3A	.5616	.5596	.938	.928	.723	.562	.279	.190	.4410	.4380	.203			
MS20010	5/8 -18 UNF-3A	.6240	.6220	1.050	1.040	.826	.625	.317	.216	.5035	.5005	.234			.019
MS20012	3/4 -16 UNF-3A	.7488	.7468	1.230	1.220	.921	.750	.385	.273	.5660	.5630	.281			
MS20014	7/8 -14 UNF-3A	.8737	.8707	1.438	1.428	1.083	.875	.450	.327	.6295	.6257	.344	.094	.055	
MS20016(e)	1 -14 NS -3A	.9985	.9955	1.625	1.615	1.249	1.000	.527	.381	.7547	.7507	.391			
MS20017	1 -12 UNF-3A														
MS20018	1-1/8 -12 UNF-3A	1.124	1.121	1.875	1.865	1.275	1.125	.589	.443			.453			.022
MS20020	1-1/4 -12 UNF-3A	1.249	1.246	2.125	2.115	1.598	1.250	.667	.475	1.005	1.001	.500			
MS20022	1-3/8 -12 UNF-3A	1.374	1.370	2.313	2.303	1.768	1.375	.746	.531	1.131	1.126	.547	.125		
MS20024	1-1/2 -12 UNF-3A	1.499	1.495	2.500	2.490	1.931	1.500	.823	.586	1.256	1.251	.594	.156		
MS20012A (D)	3/4 -16 UNF-3A	.7488	.7468	1.230	1.220	.921	.750	.385	.273	.5660	.5630	.281	.094		

TABLE I (CONT'D)

BASIC MS PART NO.	P RAD MAX	P RAD MIN	R RAD MAX	R RAD MIN	(b) S MIN REF	U ±.016	(c) V MAX	(c) W MAX	(b)(c) CENTER DRILL SIZE REF	TENSILE STRENGTH LB MIN	(a) CONCENTRICITY X	Y	Z
MS20004			.041	.031	.475	.031				6,190	.005	.007	
MS20005					.537					9,820	.006	.008	.0045
MS20006					.662	.047				15,200	.008	.010	
MS20007	.020	.010	.057	.047	.787					20,600	.009	.011	
MS20008										27,400	.010	.012	
MS20009					.850					34,800	.011	.014	.006
MS20010					.912	.062				43,600	.012	.016	
MS20012					1.037					63,200	.015	.018	
MS20014			.073	.063	1.162		.187	.200	2	86,100	.018	.022	
MS20016(e)						.078				112,000			
MS20017	.030	.015			1.287		.250	.260	3	114,000	.020	.025	.009
MS20018					1.475					144,000	.022	.026	
MS20020					1.600					180,000	.025	.032	
MS20022			.089	.077	1.725	.094	.312	.320	4	219,000	.028	.035	.012
MS20024					1.850					263,000	.030	.039	
MS20012A			.073	.063	.730	.062	.187	.200	2	63,200	.015	.018	.006 (D)

APPROVED 28 Feb 52 REVISED (D) FOR CHANGES SEE SHEETS 2 AND 4

P.A. Navy -- AS
Other Cust
Army - AR
USAF - 99

TITLE
BOLTS, INTERNAL WRENCHING, 160 KSI FTU AND 96 KSI FSU

MILITARY STANDARD
MS 20004 THRU 20024

PROCUREMENT SPECIFICATION
MIL-B-7838

SUPERSEDES:

SHEET 2 OF 4

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TABLE II DASH NOS, GRIP LENGTHS AND BOLT LENGTH

DASH NO.	GRIP ±.010	L, LENGTH ±.015														
		MS20004 1/4	MS20005 5/16	MS20006 3/8	MS20007 7/16	MS20008 1/2	MS20009 9/16	MS20010 5/8	MS20012 3/4	MS20014 7/8	MS20016 1	MS20017 1	MS20018 1-1/8	MS20020 1-1/4	MS20022 1-3/8	MS20024 1-1/2
4	.250	.750														
6	.375	.875	.938	1.062												
8	.500	1.000	1.062	1.188	1.312	1.312										
10	.625	1.125	1.188	1.312	1.438	1.438	1.500	1.562								
12	.750	1.250	1.312	1.438	1.562	1.562	1.625	1.688	1.812							
14	.875	1.375	1.438	1.562	1.688	1.688	1.750	1.812	1.938	2.062						
16	1.000	1.500	1.562	1.688	1.812	1.812	1.875	1.938	2.062	2.188	2.312	2.312	2.625			
18	1.125	1.625	1.688	1.812	1.938	1.938	2.000	2.062	2.188	2.312	2.438	2.438				
20	1.250	1.750	1.812	1.938	2.062	2.062	2.125	2.188	2.312	2.438	2.562	2.562	2.750	2.875		
22	1.375	1.875	1.938	2.062	2.188	2.188	2.250	2.312	2.438	2.562	2.688	2.688	2.875	3.000	3.125	
24	1.500	2.000	2.062	2.188	2.312	2.312	2.375	2.438	2.562	2.688	2.812	2.812	3.000	3.125	3.250	3.375
26	1.625	2.125	2.188	2.312	2.438	2.438	2.500	2.562	2.688	2.812	2.938	2.938	3.125	3.250	3.375	3.500
28	1.750	2.250	2.312	2.438	2.562	2.562	2.625	2.688	2.812	2.938	3.062	3.062	3.250	3.375	3.500	3.625
30	1.875	2.375	2.438	2.562	2.688	2.688	2.750	2.812	2.938	3.062	3.188	3.188	3.375	3.500	3.625	3.750
32	2.000	2.500	2.562	2.688	2.812	2.812	2.875	2.938	3.062	3.188	3.312	3.312	3.500	3.625	3.750	3.875
34	2.125	2.625	2.688	2.812	2.938	2.938	3.000	3.062	3.188	3.312	3.438	3.438	3.625	3.750	3.875	4.000
36	2.250	2.750	2.812	2.938	3.062	3.062	3.125	3.188	3.312	3.438	3.562	3.562	3.750	3.875	4.000	4.125
38	2.375	2.875	2.938	3.062	3.188	3.188	3.250	3.312	3.438	3.562	3.688	3.688	3.875	4.000	4.125	4.250
40	2.500	3.000	3.062	3.188	3.312	3.312	3.375	3.438	3.562	3.688	3.812	3.812	4.000	4.125	4.250	4.375
42	2.625	3.125	3.188	3.312	3.438	3.438	3.500	3.562	3.688	3.812	3.938	3.938	4.125	4.250	4.375	4.500
44	2.750	3.250	3.312	3.438	3.562	3.562	3.625	3.688	3.812	3.938	4.062	4.062	4.250	4.375	4.500	4.625
46	2.875	3.375	3.438	3.562	3.688	3.688	3.750	3.812	3.938	4.062	4.188	4.188	4.375	4.500	4.625	4.750
48	3.000	3.500	3.562	3.688	3.812	3.812	3.875	3.938	4.062	4.188	4.312	4.312	4.500	4.625	4.750	4.875
50	3.125	3.625	3.688	3.812	3.938	3.938	4.000	4.062	4.188	4.312	4.438	4.438	4.625	4.750	4.875	5.000
52	3.250	3.750	3.812	3.938	4.062	4.062	4.125	4.188	4.312	4.438	4.562	4.562	4.750	4.875	5.000	5.125
54	3.375	3.875	3.938	4.062	4.188	4.188	4.250	4.312	4.438	4.562	4.688	4.688	4.875	5.000	5.125	5.250
56	3.500	4.000	4.062	4.188	4.312	4.312	4.375	4.438	4.562	4.688	4.812	4.812	5.000	5.125	5.250	5.375
58	3.625	4.125	4.188	4.312	4.438	4.438	4.500	4.562	4.688	4.812	4.938	4.938	5.125	5.250	5.375	5.500
60	3.750	4.250	4.312	4.438	4.562	4.562	4.625	4.688	4.812	4.938	5.062	5.062	5.250	5.375	5.500	5.625
62	3.875	4.375	4.438	4.562	4.688	4.688	4.750	4.812	4.938	5.062	5.188	5.188	5.375	5.500	5.625	5.750
64	4.000	4.500	4.562	4.688	4.812	4.812	4.875	4.938	5.062	5.188	5.312	5.312	5.500	5.625	5.750	5.875
66	4.125	4.625	4.688	4.812	4.938	4.938	5.000	5.062	5.188	5.312	5.438	5.438	5.625	5.750	5.875	6.000
68	4.250	4.750	4.812	4.938	5.062	5.062	5.125	5.188	5.312	5.438	5.562	5.562	5.750	5.875	6.000	6.125
70	4.375	4.875	4.938	5.062	5.188	5.188	5.250	5.312	5.438	5.562	5.688	5.688	5.875	6.000	6.125	6.250
72	4.500	5.000	5.062	5.188	5.312	5.312	5.375	5.438	5.562	5.688	5.812	5.812	6.000	6.125	6.250	6.375
74	4.625	5.125	5.188	5.312	5.438	5.438	5.500	5.562	5.688	5.812	5.938	5.938	6.125	6.250	6.375	6.500
76	4.750	5.250	5.312	5.438	5.562	5.562	5.625	5.688	5.812	5.938	6.062	6.062	6.250	6.375	6.500	6.625
78	4.875	5.375	5.438	5.562	5.688	5.688	5.750	5.812	5.938	6.062	6.188	6.188	6.375	6.500	6.625	6.750
80	5.000	5.500	5.562	5.688	5.812	5.812	5.875	5.938	6.062	6.188	6.312	6.312	6.500	6.625	6.750	6.875
82	5.125	5.625	5.688	5.812	5.938	5.938	6.000	6.062	6.188	6.312	6.438	6.438	6.625	6.750	6.875	7.000
84	5.250	5.750	5.812	5.938	6.062	6.062	6.125	6.188	6.312	6.438	6.562	6.562	6.750	6.875	7.000	7.125
86	5.375	5.875	5.938	6.062	6.188	6.188	6.250	6.312	6.438	6.562	6.688	6.688	6.875	7.000	7.125	7.250
88	5.500	6.000	6.062	6.188	6.312	6.312	6.375	6.438	6.562	6.688	6.812	6.812	7.000	7.125	7.250	7.375
90	5.625	6.125	6.188	6.312	6.438	6.438	6.500	6.562	6.688	6.812	6.938	6.938	7.125	7.250	7.375	7.500
92	5.750	6.250	6.312	6.438	6.562	6.562	6.625	6.688	6.812	6.938	7.062	7.062	7.250	7.375	7.500	7.625
94	5.875	6.375	6.438	6.562	6.688	6.688	6.750	6.812	6.938	7.062	7.188	7.188	7.375	7.500	7.625	7.750
96	6.000	6.500	6.562	6.688	6.812	6.812	6.875	6.938	7.062	7.188	7.312	7.312	7.500	7.625	7.750	7.875
98	6.125						7.000	7.062	7.188	7.312	7.438	7.438	7.625	7.750	7.875	8.000
100	6.250						7.125	7.188	7.312	7.438	7.562	7.562	7.750	7.875	8.000	8.125
102	6.375						7.250	7.312	7.438	7.562	7.688	7.688	7.875	8.000	8.125	8.250
104	6.500						7.375	7.438	7.562	7.688	7.812	7.812	8.000	8.125	8.250	8.375
106	6.625						7.500	7.562	7.688	7.812	7.938	7.938	8.125	8.250	8.375	8.500
108	6.750						7.625	7.688	7.812	7.938	8.062	8.062	8.250	8.375	8.500	8.625
110	6.875						7.750	7.812	7.938	8.062	8.188	8.188	8.375	8.500	8.625	8.750
112	7.000						7.875	7.938	8.062	8.188	8.312	8.312	8.500	8.625	8.750	8.875
114	7.125												8.625	8.750	8.875	9.000
116	7.250												8.750	8.875	9.000	9.125
118	7.375												8.875	9.000	9.125	9.250
120	7.500												9.000	9.125	9.250	9.375
122	7.625												9.125	9.250	9.375	9.500
124	7.750												9.250	9.375	9.500	9.625
126	7.875												9.375	9.500	9.625	9.750
128	8.000												9.500	9.625	9.750	9.875

UNASSIGNED DASH NOS. SHALL NOT BE USED, EXCEPT THOSE DASH NOS. COVERED BY THE FOLLOWING CODING NOTE.
DASH NUMBERS INDICATE PREFERRED GRIP LENGTHS IN .125 INCREMENTS. INTERMEDIATE GRIP LENGTHS BETWEEN THOSE LENGTHS LISTED, MAY BE SPECIFIED IN .062 INCREMENTS BY INTERMEDIATE DASH NUMBERS. LENGTH AND GRIP OF A BOLT WITH AN ODD NUMBER FOR THE DASH NUMBER WILL BE .062 GREATER THAN SHOWN FOR THE BOLT WITH THE PRECEDING EVEN DASH NUMBER. ONLY BOLT DIAMETER FOR WHICH THERE ARE QUALIFIED PRODUCTS LISTED ON QPL-7838 SHALL BE USED FOR DESIGN.
FOR EACH BOLT DIAMETER GRIP LENGTHS GREATER OR LESS THAN THOSE DESIGNATED BY ASSIGNED DASH NUMBERS ON SHEET 2 ARE INACTIVE FOR DESIGN.
ADD H BEFORE DASH NO. FOR BOLT WITH DRILLED HEAD.
EXAMPLES OF PART NOS.: MS20017-18 = 1-12 BOLT, 2.438 INCH LONG, 1.125 INCH GRIP, UNDRILLED HEAD.
MS20017H18 = 1-12 BOLT, 2.438 INCH LONG, 1.125 INCH GRIP, DRILLED HEAD.

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REVIEWER SYMBOLS: ARMY-ER ME DLA-IS

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APPROVED 28 Jan 72 REVISED (2) FOR CHANGES SEE SHEETS 2 AND 4

P.A. Navy - AS Other Cust Army - AR USAF - 99	TITLE BOLT - INTERNAL WRENCHING, 160 KSI FTU AND 96 KSI FSU	MILITARY STANDARD MS20004 THRU MS20024
PROCUREMENT SPECIFICATION MIL-B-7838	SUPERSEDES:	SHEET 3 OF 4

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TABLE II (CONT'D)

DASH NO	GRIP ±.010	L, LENGTH, T.015 MS20012A 3/4
8	.500	1.230

(D)

APPROVED 28 Feb. 1952 REVISED (D) FOR CHANGES SEE SHEETS 2 AND 4

P.A. NAVY - AS Other Cust Army - AR USAF - 99	TITLE BOLTS, INTERNAL WRENCHING, 160 KSI FTU AND 96 KSI FSU	MILITARY STANDARD MS 20004 THRU 20024
PROCUREMENT SPECIFICATION MIL-B-7838	SUPERSEDES:	SHEET 4 OF 4

DD FORM 1 SEP 71 672-1 (Coordinated)

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